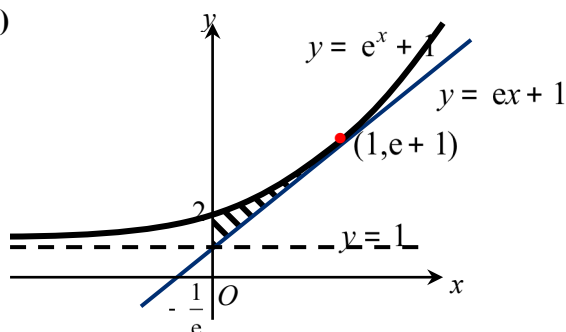


Solution

Section A

$$\left. \begin{aligned} 1 \quad & 2^x = 2^2(2^{3y}); \\ & x = 2 + 3y \text{ -- (1)} \\ & \log_5 \frac{x}{x-4y} = 1 \\ & \frac{x}{x-4y} = 5 \\ & x = 5y \text{ -- (2)} \end{aligned} \right\} \begin{array}{l} \text{equate (1) \& (2), } 5y = 2 + 3y \\ \qquad \qquad \qquad y = 1 \text{ and } x = 5 \end{array}$$

2(i)



(ii) Area of shaded region

$$= \hat{U}_0^1(e^x + 1) - (ex + 1) dx = \hat{U}_0^1(e^x - ex) dx$$

$$= \left[e^x - \frac{ex^2}{2} \right]_0^1 = \left[e^1 - \frac{e(1)^2}{2} \right] - \left[e^0 - \frac{e(0)^2}{2} \right] = \frac{e}{2} - 1$$

3 Given $AD = x$ cm, let $AB = y$ cm. $\setminus 3x + 2y = 24$

$$\text{Height of } \triangle ADE = \sqrt{x^2 - \left(\frac{\hat{E}_1 - \hat{E}_2}{4}\right)^2} = \sqrt{\frac{3}{4}x^2} = \frac{\sqrt{3}}{2}x$$

$$\left. \begin{aligned} S &= xy + \frac{1}{2}x\sqrt{3} - \frac{3x^2}{2} = 12x - \frac{3x^2}{2} + \frac{\sqrt{3}}{4}x^2 \\ &= 12x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2 = 12x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2 \quad (\text{shown}) \end{aligned} \right\} \begin{aligned} \frac{dS}{dx} &= 12 + 2\left(\frac{\sqrt{3}}{4} - \frac{3}{2}\right)x. \text{ At stationary points, } \frac{dS}{dx} = 0 \\ \text{Since } S &\text{ is a quadratic function of } x \text{ with} \\ \therefore x &\approx 5.6233 = 5.62 \text{ (3 s.f.)} \\ \text{coefficient of } x^2 &\left(\frac{\sqrt{3}}{4} - \frac{3}{2}\right) < 0, S \text{ has a} \\ &\text{maximum value.} \end{aligned}$$

$\therefore$ Maximum value of $S \approx 33.7398 = 33.74$ (2 d.p.)

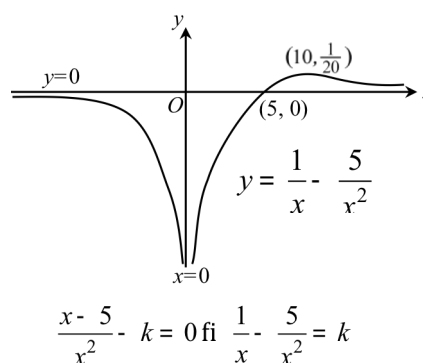
$$\text{4(i)} \quad y = \frac{1}{x} - \frac{5}{x^2} \text{ fi } \frac{dy}{dx} = -\frac{1}{x^2} + \frac{10}{x^3}$$
$$\frac{dy}{dx} = 0 \text{ fi } \frac{10}{x^3} = \frac{1}{x^2}$$
$$\setminus x = 10 \text{ (Q x π 0)}$$
$$\frac{dy}{dx} = 0 \text{ for exactly one value of } x. (\text{shown})$$

Corresponding value of $y = \frac{1}{10} - \frac{5}{100} = \frac{1}{20}$.

$$\frac{x-5}{x^2} - k = 0 \text{ has no real roots} \Rightarrow \text{line } y = k \text{ does not cut the above graph. } \setminus k > \frac{1}{20}.$$

(iii) When $x = 5$, $\frac{dy}{dx} = -\frac{1}{5^2} + \frac{10}{5^3} = \frac{1}{25}$. Equation of normal at (5,0) is $y - 0 = -25(x - 5)$ i.e. $y = 125 - 25x$.

(ii)


$$\frac{x-5}{x^2} - k = 0 \text{ fi } \frac{1}{x} - \frac{5}{x^2} = k$$

e graph. $\setminus k > \frac{1}{20}$.

$$\begin{aligned} 5(i)(a) \quad \frac{d}{dx} \frac{3}{(5-x)^2} &= -\frac{3}{(5-x)^2}(-1) \\ &= \frac{3}{(5-x)^2} \end{aligned}$$

$$\begin{aligned} (b) \quad y &= \ln(2x^2 + 4)^3 \\ &= 3 \ln(2x^2 + 4) \\ \frac{dy}{dx} &= \frac{3}{2x^2 + 4}(4x) \\ &= \frac{6x}{x^2 + 2} \end{aligned}$$

$$\begin{aligned} (ii) \quad \int_1^8 (x^{\frac{1}{3}} + x^{-\frac{1}{3}})^2 dx &= \int_1^8 (x^{\frac{2}{3}} + 2 + x^{-\frac{2}{3}}) dx \\ &= \left[\frac{3}{5} x^{\frac{5}{3}} + 2x + 3x^{\frac{1}{3}} \right]_1^8 \\ &= \frac{3}{5}(\sqrt[3]{8})^5 - 1 + 2(8 - 1) + 3(\sqrt[3]{8} - 1) \\ &= \frac{3}{5}(32 - 1) + 17 \\ &= \frac{178}{5} = 35\frac{3}{5} \end{aligned}$$

Section B

- 6 (i) Each member in the record list is given a distinct number from 1 to 400. Use a random number generator to generate a number from 1 to 5. The member whose number is generated will be selected. Starting from this member, every $\frac{400}{80} = 5^{\text{th}}$ member in the list will be selected until 80 members are obtained.

- (ii) Systematic sampling may result in a non-representative sample, for example, very few members from the swimming section may be selected hence those surveyed may not be able to comment on the swimming facilities.

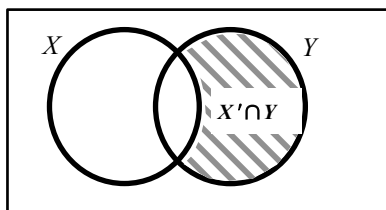
- (iii) Use stratified sampling.

- Divide the population into strata by sections and calculate the number of members to be surveyed within each stratum as follows.

	Male	Female
Badminton	$\frac{105}{400} \times 80 = 21$	$\frac{60}{400} \times 80 = 12$
Swimming	$\frac{90}{400} \times 80 = 18$	$\frac{45}{400} \times 80 = 9$
Tennis	$\frac{45}{400} \times 80 = 9$	$\frac{55}{400} \times 80 = 11$

- Obtain the name list of all members in the badminton section. Use a random number generator to generate 33 distinct numbers and members associate with these numbers will be selected. Repeat the process for the remaining 2 sections.

7(a)(i)

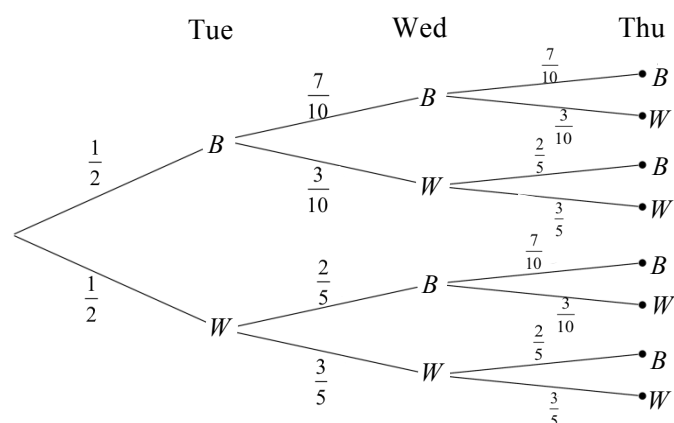


$$P(X \cap Y) = P(Y) - P(X' \cap Y) = \frac{1}{2} - \frac{1}{2} = 0.$$

$\therefore X$ and Y are mutually exclusive events.

$$\begin{aligned} (ii) \quad P(X' \cap Y) &= 1 - P(X \cup Y) \\ &= 1 - P(X) - P(Y) \\ &= 1 - \frac{1}{3} - \frac{1}{2} \\ &= \frac{1}{6} \end{aligned}$$

- 7(b)** Let B denote the event that the boy goes to school by bus on one day and W denote the event that he walks to school on one day.



$P(\text{goes by bus on Thurs} | \text{walks on Tues})$

$$= \frac{2}{5} \times \frac{7}{10} + \frac{3}{5} \times \frac{2}{5} = \frac{7}{25} + \frac{6}{25} = \frac{13}{25}$$

$P(\text{also walks on Wed} | \text{walks on Tues and Thurs})$

$$= \frac{P(\text{walks on all 3 days})}{P(\text{walks on Tues and Thurs})}$$

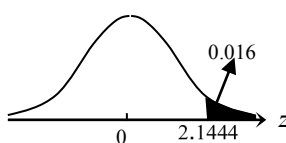
$$= \frac{\frac{1}{2} \times \frac{3}{5} \times \frac{3}{5}}{\frac{1}{2} \times \frac{2}{5} \times \frac{3}{10} + \frac{1}{2} \times \frac{3}{5} \times \frac{3}{5}} = \frac{9}{25} \times \frac{25}{12} = \frac{3}{4}$$

- 8** Let A cm and B cm be the diameter of a circular disc made by machine A and machine B respectively.

(i) $A \sim N(8.95, \sigma^2)$

$$P(A > 9.02) = 0.016$$

$$P\left(Z > \frac{9.02 - 8.95}{\sigma}\right) = 0.016$$



$$\therefore \frac{0.07}{\sigma} = 2.1444$$

$$\Rightarrow \sigma \approx 0.032643 = 0.033 \text{ (3 d.p.)}$$

- (ii) $E(B) = 9, \text{Var}(B) = 0.15^2$. Since n is sufficiently large, by Central Limit Theorem,

$$\bar{B} = \frac{B_1 + \dots + B_{50}}{50} \sim N\left(9, \frac{0.15^2}{50}\right) \text{ approximately. } \therefore P(\bar{B} > 9.05) = 0.00921$$

(iii) $A \sim N(8.95, 0.032643^2), B \sim N(9, 0.15^2)$

$$P(A_1 + A_2 + A_3 < 3B) = P(A_1 + A_2 + A_3 - 3B < 0)$$

$$E(A_1 + A_2 + A_3 - 3B) = 3E(A) - 3E(B) = -0.15$$

$$\therefore A_1 + A_2 + A_3 - 3B \sim N(-0.15, 0.205697)$$

$$\text{Var}(A_1 + A_2 + A_3 - 3B) = 3\text{Var}(A) + 3^2\text{Var}(B) = 0.205697$$

$$P(A_1 + A_2 + A_3 - 3B < 0) = 0.62958 \approx 0.630 \text{ (3 s.f.)}$$

- 9** unbiased estimate of the population mean, μ

$$= \bar{x} = \frac{\sum x}{n} = \frac{558.2}{100} = 5.582$$

- unbiased estimate of the population variance, σ^2

$$= s^2 = \frac{1}{n-1} \left[\sum x^2 - \frac{(\sum x)^2}{n} \right] = \frac{1}{99} \left[4027.89 - \frac{(558.2)^2}{100} \right]$$

$$\approx 9.2123 = 9.21 \text{ (3 s.f.)}$$

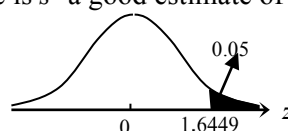
(i) $H_0: \mu = 5$

$$H_1: \mu > 5$$

Level of significance: 5%

Test Statistic: $\bar{X} : N\left(5, \frac{\sigma^2}{n}\right)$ when H_0 is true. Since s^2 is a good estimate of σ^2 , $Z = \frac{\bar{X} - 5}{s/\sqrt{n}} : N(0,1)$

Computation: $\bar{x} = 5.582, s^2 = 9.2123, n = 100$
 $z = 1.9175, p = 0.027586$



Conclusion: Since p -value $\approx 0.0276 < 0.05$, H_0 is rejected in favour of H_1 at 5% significance level. Hence there is sufficient evidence to conclude that the mean time spent shopping by customers is more than 5 hours at the 5% significance level.

- (ii) (a) Test is still valid. Since $n = 100$ is sufficiently large, by Central Limit Theorem, the sample mean time spent by the 100 shoppers ($\bar{X}$) is approximately normal.

(b) Test is not valid as the time spent shopping at weekend may be biased by different lifestyles.

(iii) $H_0 : \mu = \mu_0$

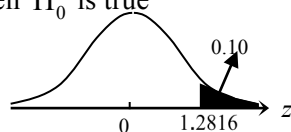
$H_1 : \mu > \mu_0$

Level of significance: 10%

Test statistic: $Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} : N(0,1)$ when H_0 is true

Rejection region: $z \geq 1.2816$

Computation: $z = \frac{5.582 - \mu_0}{\sqrt{\frac{9.2123}{100}}}$



$$H_0 \text{ is not rejected} \Rightarrow \frac{5.582 - \mu_0}{\sqrt{\frac{9.2123}{100}}} < 1.2816 \Rightarrow 5.582 - \mu_0 < 1.2816 \sqrt{\frac{9.2123}{100}} \Rightarrow \mu_0 > 5.1930$$

$\therefore$ least value of $\mu_0 = 5.2$ (1 d.p)

10 (i) $X \sim B(50, 0.83)$ Assumptions:

- Passengers turn up independently of each other for all the flights.
- The probability that a passenger turns up for a flight remains constant for all the passengers of the flight.

(ii) $P(X > 45) = 1 - P(X \leq 45) = 0.057226 \approx 0.0572$

(iii) Let Y be the number of flights out of 15 with passengers without seats. $Y \sim B(15, 0.057226)$

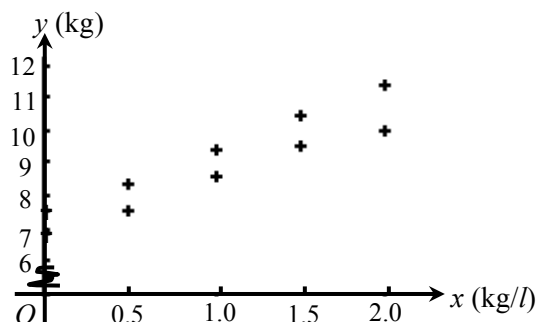
$P(1 \leq Y < 4) = P(Y \leq 3) - P(Y = 0) = 0.578$

(iv) Let W be the number of flights out of 100 with passengers without seats. $W \sim B(100, 0.057226)$

Since n is large & $np = 5.7226 > 5$, $n(1-p) = 94.3 > 5$, $W \sim N(5.7226, 5.3951)$ approx.

$P(5 < W \leq 12) = P(5.5 \leq W \leq 12.5) = 0.536$ (3 s.f.)

11(i)



(ii) The product moment correlation coefficient between x and y , $r \approx 0.93308 \approx 0.933$.

The value of r is close to +1 and the points in the scatter diagram seem to lie close to an ascending straight line are indications of a strong positive linear relationship between concentration of fertiliser (x) and crop yield (y). This means that as x increases, y tends to increase at a constant rate.

(iii) Equation of the regression line of y on x is $y = 1.79x + 7.18$.

The gradient of the line, $m = 1.79$ means that the expected increase in crop yield is 1.79 kg for every additional concentration of 1 kg per litre of fertiliser used in one of the plots.

When no fertiliser is used, the expected crop yield, c is 7.18 kg.

(iv) The line is not suitable for extrapolation as $x = 2.5$ lies outside the range of $0 \leq x \leq 2.0$.

Since the levels of concentration of fertiliser (x) are pre-determined, so x is not random (no error/variation). Hence only the regression line of y on x is meaningful. Therefore, the regression line of y on x will be used for estimation of x given y .

When $y = 10$, $x = \frac{10 - 7.18}{1.79} \approx 1.5754 \approx 1.58$ kg/litre.

Since the linear model is appropriate (from (ii)) and $y = 10$ is within the data range of $6.8 \leq y \leq 11.4$, the estimate is reliable.